

Mathematics Questions - Algebra

Subject: Mathematics | Chapter: Algebra | Difficulty: medium

Question 1: Solve the equation $2x^2 + 5x - 3 = 0$

Options:

- A) $x = 1/2, x = -3$
- B) $x = -1/2, x = 3$
- C) $x = 1, x = -3/2$
- D) $x = -1, x = 3/2$

Difficulty: 4/5

Question 2: Simplify the expression $(3x^2y^3)^2(2xy t)^3$

Options:

- A) $108x wy^1 x$
- B) $108x uy^1 x$
- C) $72x wy^1 x$
- D) $72x uy^1 x$

Difficulty: 4/5

Question 3: Solve the system of equations: $x + y = 5$ and $x - y = 1$

Options:

- A) $x = 2, y = 3$
- B) $x = 3, y = 2$
- C) $x = 4, y = 1$
- D) $x = 1, y = 4$

Difficulty: 3/5

Question 4: What is the slope of the line passing through points (2, 5) and (4, 9)?

Options:

- A) 2
- B) $\frac{1}{2}$
- C) -2
- D) $-\frac{1}{2}$

Difficulty: 3/5

Question 5: If $f(x) = 2x + 3$, find $f^{-1}(x)$

Options:

- A) $f^{-1}(x) = (x - 3)/2$
- B) $f^{-1}(x) = (x + 3)/2$
- C) $f^{-1}(x) = 2x - 3$
- D) $f^{-1}(x) = 3x - 2$

Difficulty: 4/5

Question 6: Expand $(x + 2)(x^2 - 2x + 4)$

Options:

- A) $x^3 + 8$
- B) $x^3 - 8$
- C) $x^3 + 4x^2 - 8$
- D) $x^3 - 4x^2 + 8$

Difficulty: 4/5

Question 7: Solve the inequality $3x - 6 > 9$

Options:

- A) $x > 5$
- B) $x < 5$
- C) $x > -5$
- D) $x < -5$

Difficulty: 3/5

Question 8: Find the discriminant of the quadratic equation $x^2 - 4x + 4 = 0$

Options:

- A) 0
- B) 16
- C) -16
- D) 4

Difficulty: 3/5

Question 9: Simplify the expression $75 + 12$

Options:

- A) 73
- B) 53
- C) 37
- D) 87

Difficulty: 4/5

Question 10: What are the x-intercepts of the quadratic function $y = x^2 - 5x + 6$?

Options:

- A) $x = 2, x = 3$
- B) $x = -2, x = -3$
- C) $x = 2, x = -3$
- D) $x = -2, x = 3$

Difficulty: 4/5

Generated on: 30/7/2025, 3:13:00 am